

Q1 - 24 June - Shift 1

A block of mass 10 kg starts sliding on a surface with an initial velocity of 9.8 ms^{-1} . The coefficient of friction between the surface and block is 0.5. The distance covered by the block before coming to rest is : [use $g = 9.8 \text{ ms}^{-2}$]

- (A) 4.9 m (B) 9.8 m
(C) 12.5 m (D) 19.6 m

Space for your notes:

Q2 - 24 June - Shift 1

A boy ties a stone of mass 100 g to the end of a 2 m long string and whirls it around in a horizontal plane. The string can withstand the maximum tension of 80 N. If the maximum speed with which

the stone can revolve is $\frac{K}{\pi} \text{ rev./min}$. The value of

K is : (Assume the string is massless and unstretchable)

- (A) 400 (B) 300
(C) 600 (D) 800

Space for your notes:

Q3 - 24 June - Shift 1

Questions

MathonGo

A particle experiences a variable force

$\vec{F} = (4x\hat{i} + 3y^2\hat{j})$ in a horizontal x-y plane. Assume

distance in meters and force is newton. If the

particle moves from point (1, 2) to point (2, 3) in

the x-y plane, the Kinetic Energy changes by

(A) 50.0 J

(B) 12.5 J

(C) 25.0 J

(D) 0 J

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Q4 - 24 June - Shift 1

A ball of mass 100 g is dropped from a height $h =$

10 cm on a platform fixed at the top of

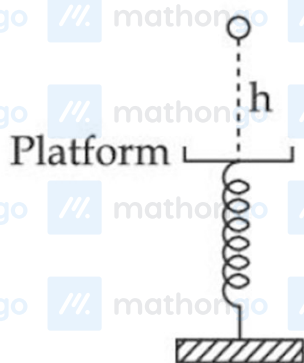
vertical spring (as shown in figure). The ball stays

on the platform and the platform is depressed by a

distance $\frac{h}{2}$. The spring constant is _____ Nm^{-1} .

(Use $g = 10 \text{ ms}^{-2}$)

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Q5 - 24 June - Shift 2

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Questions

MathonGo

A stone of mass m , tied to a string is being whirled in a vertical circle with a uniform speed. The tension in the string is :

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- (A) the same throughout the motion
- (B) minimum at the highest position of the circular path
- (C) minimum at the lowest position of the circular path
- (D) minimum when the rope is in the horizontal position

Q6 - 24 June - Shift 2

Potential energy as a function of r is given by

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$U = \frac{A}{r^{10}} - \frac{B}{r^5}$, where r is the interatomic distance,

A and B are positive constants. The equilibrium distance between the two atoms will be :

- (A) $\left(\frac{A}{B}\right)^{\frac{1}{5}}$
- (B) $\left(\frac{B}{A}\right)^{\frac{1}{5}}$
- (C) $\left(\frac{2A}{B}\right)^{\frac{1}{5}}$
- (D) $\left(\frac{B}{2A}\right)^{\frac{1}{5}}$

Q7 - 25 June - Shift 1

A 0.5 kg block moving at a speed of 12 ms^{-1} compresses a spring through a distance 30 cm when its speed is halved. The spring constant of the spring will be _____ Nm^{-1} .

*Space for your notes:***Q8 - 25 June - Shift 2**

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Questions

MathonGo

The ratio of specific heats $\left(\frac{C_P}{C_V}\right)$ in terms of degree of freedom (f) is given by:

(A) $\left(1 + \frac{f}{3}\right)$ (B) $\left(1 + \frac{2}{f}\right)$

(C) $\left(1 + \frac{f}{2}\right)$ (D) $\left(1 + \frac{1}{f}\right)$

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Q9 - 26 June - Shift 2

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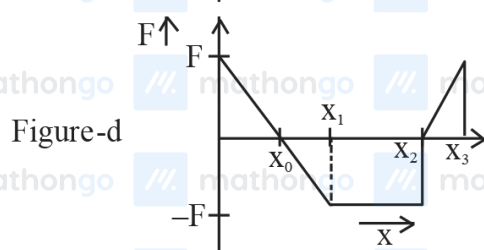
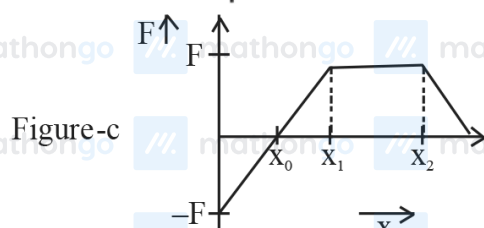
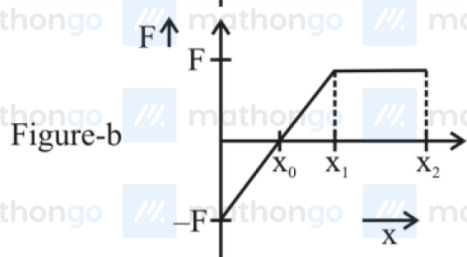
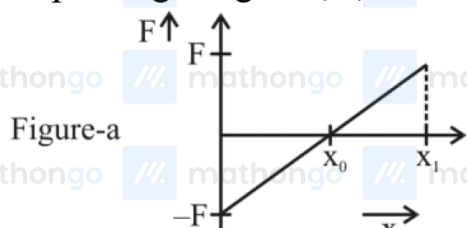
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Questions

MathonGo

Arrange the four graphs in descending order of total work done; where W_1 , W_2 , W_3 and W_4 are the work done corresponding to figure a, b, c and d respectively.

Space for your notes:



- (A) $W_3 > W_2 > W_1 > W_4$
 (B) $W_3 > W_2 > W_4 > W_1$
 (C) $W_2 > W_3 > W_4 > W_1$
 (D) $W_2 > W_3 > W_1 > W_4$

Q10 - 27 June - Shift 2

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Questions

MathonGo

A stone tied to a string of length L is whirled in a vertical circle with the other end of the string at the centre. At a certain instant of time, the stone is at its lowest position and has a speed u . The magnitude of change in its velocity, as it reaches a position where the string is horizontal, is $\sqrt{x(u^2 - gL)}$. The value of x is

(A) 3

(B) 2

(C) 1

(D) 5

*Space for your notes:***Q11 - 27 June - Shift 2**

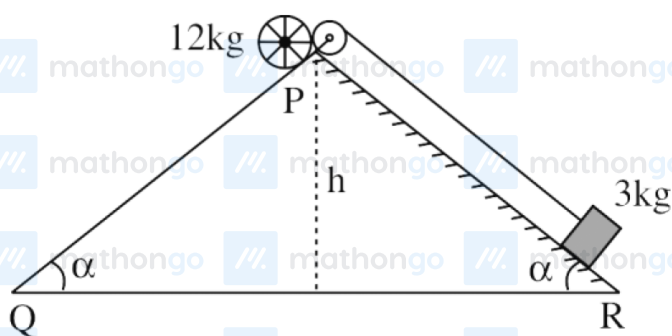
#MathBoleTohMathonGo

Questions

MathonGo

A rolling wheel of 12 kg is on an inclined plane at position P and connected to a mass of 3 kg through a string of fixed length and pulley as shown in figure. Consider PR as friction free surface.

The velocity of centre of mass of the wheel when it reaches at the bottom Q of the inclined plane PQ will be $\frac{1}{2}\sqrt{xgh}$ m/s. The value of x is _____.

Space for your notes:**Q12 - 28 June - Shift 1**

A particle of mass m is moving in a circular path of constant radius r such that its centripetal acceleration (a) is varying with time t as $a = k^2 r t^2$, where k is a constant. The power delivered to the particle by the force acting on it is given as

- (A) zero
- (B) $mk^2 r^2 t^2$
- (C) $mk^2 r^2 t$
- (D) $mk^2 r t$

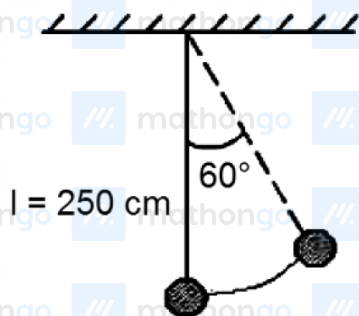
*Space for your notes:***Q13 - 28 June - Shift 1**

#MathBoleTohMathonGo

Questions

MathonGo

A pendulum is suspended by a string of length 250 cm. The mass of the bob of the pendulum is 200 g. The bob is pulled aside until the string is at 60° with vertical as shown in the figure. After releasing the bob, the maximum velocity attained by the bob will be _____ ms^{-1} . (if $g = 10 \text{ m/s}^2$)



Space for your notes:

Q14 - 29 June - Shift 1

A particle of mass 500 gm is moving in a straight line with velocity $v = b x^{5/2}$. The work done by the net force during its displacement from $x = 0$ to $x = 4 \text{ m}$ is : (Take $b = 0.25 \text{ m}^{-3/2} \text{ s}^{-1}$).

- (A) 2 J (B) 4 J
(C) 8 J (D) 16 J

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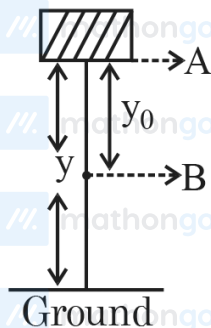
Q15 - 29 June - Shift 2

Questions

MathonGo

In the given figure, the block of mass m is dropped from the point 'A'. The expression for kinetic energy of block when it reaches point 'B' is :

Space for your notes:



(A) $\frac{1}{2}mg y_0^2$ (B) $\frac{1}{2}mg y^2$

(C) $mg(y - y_0)$ (D) $mg y_0$

Answer Key**Q1 (B)****Q2 (C)****Q3 (C)****Q4 (120)****Q5 (B)****Q6 (C)****Q7 (600)****Q8 (C)****Q9 (A)****Q10 (B)****Q11 (3)****Q12 (C)****Q13 (5)****Q14 (D)****Q15 (D)**

#MathBoleTohMathonGo

Q1 (B)

$$a = -\mu g = -0.5 \times 9.8 = -4.9 \text{ m/s}^2$$

$$d = \frac{v^2}{2a} = \frac{9.8 \times 9.8}{2(4.9)}$$

$$= 9.8 \text{ m}$$

Q2 (C)

$$T = M\omega^2 R$$

$$T = 80 \text{ N} \quad M = 0.1 \quad \omega = ? \quad R = 2 \text{ m}$$

$$80 = 0.1 \omega^2 (2)$$

$$\omega^2 = 400$$

$$\omega = 20$$

$$2\pi f = 20$$

$$f = \frac{10 \text{ rev}}{\pi \text{ s}}$$

$$= \frac{600 \text{ rev}}{\pi \text{ min}}$$

Q3 (C)

$$\vec{F} = 4x\hat{i} + 3y^2\hat{j}$$

$$WD = \Delta KE$$

$$W = \int \vec{F} \cdot (dx\hat{i} + dy\hat{j})$$

$$= \int_1^2 4x dx + \int_2^3 3y^2 dy$$

$$= (2x^2)_1^2 + (y^3)_2^3$$

$$= (8 - 2) + (27 - 8)$$

$$= 6 + 19 = 25J$$

Q4 (120)

By energy conservation

$$PE = KE$$

$$mg\left(H + \frac{H}{2}\right) = \frac{1}{2}kx^2 \quad \left(x = \frac{H}{2}\right)$$

$$0.100 \times 10 \times \frac{3}{2}(0.10) = \frac{1}{2}k(0.05 \times 0.05)$$

$$k = \frac{3 \times 0.10}{0.05 \times 0.05}$$

$$= \frac{3 \times 1000}{25} = 120 \text{ N/m}$$

Q5 (B)

Theory

Q6 (C)

$$\frac{-10A}{r^{11}} + \frac{5B}{r^6} = 0$$

$$r^5 = \frac{10A}{5B} = \frac{2A}{B}$$

Q7 (600)

$$U_i + K_i = U_f + K_f$$

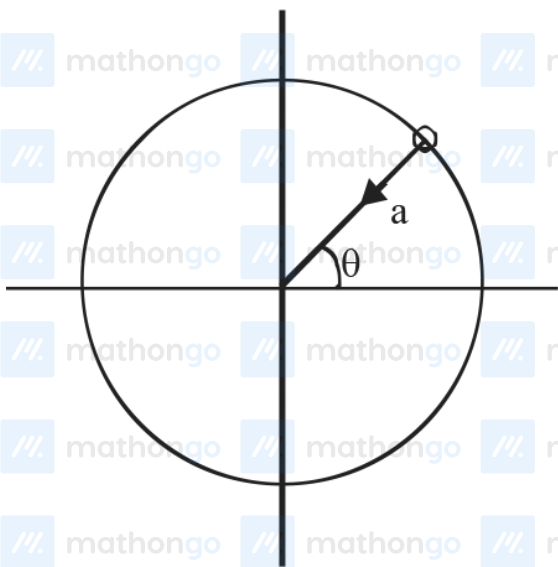
$$\Rightarrow 0 + \frac{1}{2}m(12)^2 = \frac{1}{2}K(0.3)^2 + \frac{1}{2}m(6)^2$$

$$\Rightarrow 0.5(12^2 - 6^2) = K(0.3)^2$$

$$K = 600 \text{ N/m}$$

Q8 (C)

$$a = |\vec{a}| = \frac{V^2}{R}$$



$$\vec{a} = -a \cos \theta \hat{i} - a \sin \theta \hat{j}$$

$$= -\frac{V^2}{R} \cos \theta \hat{i} - \frac{V^2}{R} \sin \theta \hat{j}$$

Q9 (A)

Work done = area under $F - x$ curve. Area below x -axis is negative & area above x -axis is positive.

so

$$W_3 > W_2 > W_1 > W_4$$

Q10 (B)

$$v = \sqrt{u^2 - 2gL}$$

$$\Delta v = \sqrt{u^2 + v^2}$$

$$\Delta v = \sqrt{u^2 + v^2 - 2gL}$$

$$\Delta v = \sqrt{2u^2 - 2gL}$$

$$\Delta v = \sqrt{2(u^2 - gL)} \quad x = 2$$

Q11 (3)

Net loss in PE = Gain in KE

$$12gh - 3gh = \frac{1}{2}3v^2 + \frac{1}{2}12v^2 + \frac{1}{2}[12r^2]\left(\frac{v}{r}\right)^2$$

$$9gh = \frac{1}{2}[3 + 12 + 12]v^2$$

$$v^2 = \frac{2gh}{3} \Rightarrow v = \frac{1}{2}\sqrt{\frac{8}{3}gh}$$

$$x = \frac{8}{3} \approx 3$$

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Q12 (C)

$$a = k^2 r t^2 = \frac{V^2}{r}$$

$$V = krt$$

$$a_t = \frac{dv}{dt} = kr$$

$$F_t = ma_t = mkr$$

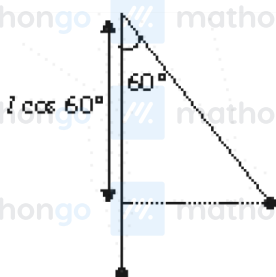
$$P = \vec{F} \cdot \vec{V}$$

$$= F \cos \theta V = F_t V = mkr(krt)$$

$$P = mk^2 r^2 t$$

Q13 (5)

$$V_{\max} = \sqrt{2gh}$$



The speed will be highest at the lowest position.

$$h = (l - l \cos 60^\circ) = \frac{l}{2}$$

$$V_{\max} = \sqrt{2 \times g \times \frac{l}{2}} = \sqrt{10 \times 2.5} = 5 \text{ m/s}$$

#MathBoleTohMathonGo

Q14 (D)

By work energy theorem

work done by net force = $\Delta K.E.$

$$\Rightarrow w = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2$$

$$w = \frac{1}{2} \times 0.5 \times (0.25)^2 \times (4)^5$$

$$w = 16J \quad (D)$$

Q15 (D)

Work done by gravity = $K_B - K_A$

$$mgy_0 = K_B - 0$$

$$K_B = mgy_0$$